The 1000-inch Wall
Samsung's new 'The Wall' is 1000-inches wide diagnally and has a resolution of 16K.
https://dgit.in/aug21-41

Zoom settles for $85mln
Zoom reaches $85 mn settlement over user privacy, 'Zoombombing'.
https://dgit.in/aug21-42

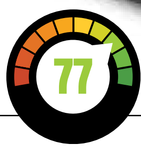
AMD Ryzen 7 5700G

Supercharged APU

AMD has been on a roll after they came out with the Zen 3-based Ryzen 5000 series desktop processors and we're now getting to see the launch of their Ryzen 5000 series APUs. APUs are generally favoured in the entry-level segment as it is targeted towards folks who'd like to save some money by not having to purchase a discrete graphics card. A high-end APU, is a slightly different story.

SPECIFICATIONS

The two new APUs fall within the mid-range and high-end segments considering their core counts. When compared to their CPU counterparts such as the Ryzen 5 5600X and Ryzen 7 5700X within the same segments, we see that these new APUs feature the same or better base clocks which is a good thing but the boost clocks are a bit lower. Having an IGP can bring down the boost clocks since they put out quite a bit of heat.



PERFORMANCE..........79
VALUE FOR MONEY....75

More interestingly, the L3 Cache for both the APUs are half of what the Vermeer processors have and even the TDP has been dropped to 65W. So we can't expect the same performance from the G variants as the X variants. As for the IGPs, the AMD Ryzen 5 5600G has 7 Radeon Compute Units with a core configuration of 448:28:8 (Shaders: Texture Mapping Units: ROPs) and the AMD Ryzen 7 5700G has 8 Radeon Compute Units giving it a core configuration of 512:32:8 (Shaders: Texture Mapping Units: ROPs). AMD's IGP will also work with AMD FSR (Fidelity Super Resolution) which is an upscaling algorithm to help boost frame rates in video games by rendering frames at lower resolutions and upscaling them. This addition will make the IGPs in all compatible AMD APUs much more powerful than what Intel can muster.

Another thing to note is that these APUs only support PCIe Gen 3.0. While gaming is not going to be affected, you will not be able to make use of the full potential of PCIe Gen 4.0 NVMe SSDs if you're using these new APUs.

AMD RYZEN 7 5700G PERFORMANCE

In the Cinebench R20 single-threaded benchmark, we see that the Ryzen 7 5700G scores 588 points which makes it the first sub-600 scorer in the Ryzen 5000 family. In the multi-threaded run, it scores 5500 which is way below the Ryzen 7 5800X's score of 5994. The Ryzen 7 5700G with its 65W TDP is nowhere close to matching the Ryzen 7 5800X despite having the same number of cores. It's higher than the Intel Core i5-11600K but that's a 6 Core / 12 Thread processor and the 5700G has two additional cores so it's not as impressive.

In POV-Ray, we see the AMD Ryzen 7 5700G perform quite similar to the older Ryzen 5 3600X. Then in Handbrake, the AMD Ryzen 5 5700G somewhat equals the Ryzen 7 3800X which is also an 8-Core processor, however, it is from the previous generation and uses the older Zen 2 microarchitecture. And in Blender, the AMD Ryzen 7 5700G with its 8 cores finally chugs ahead of the 5600X in this benchmark and even equals the 10900K.

GAMING - DISCRETE GPU

We ran the 3DMark and a few game benchmarks for the discrete graphics to see if the CPU takes a hit on account of including an IGP and we can see that the AMD Ryzen 7 5700G does indeed lose out on 13-15 per cent performance compared to the 5600X or the 5800X.

GAMING - INTEGRATED GPU

The performance of the IGP in the 5700G is about 42.5 per cent of what the GTX 1650 is capable of. Compared to the 3400G, it is about 22 per cent better. But, these are synthetic benchmark scores and actual game benchmarks portray a slightly different picture.

In competitive games such as CS:GO, Dota 2 and Fortnite, the AMD Ryzen 7 5700G scored 220, 64 and 91 FPS, respectively. Whereas games such as Shadow of the Tomb Raider and The Witcher 3 were a little taxing bring the frame rate down to 44 and 34 FPS, respectively. All tests were done at 1080p at low settings.

Finally, we switched on FSR and in some titles, the improvement could be nearly 2x as was the case in The Riftbreaker whereas, in Godfall, the bump was 19 per cent.

VERDICT

The 5700G is a pretty decent APU with its powerful IGP. That said, we were hoping for AMD to launch ebtry-level Ryzen 5000 APUs since they serve the budget segment better. But for now, these will do.

–Mithun Mohandas

SPECIFICATIONS

CORES: 8. | **THREADS:** 16 | **GPU CORES:** 8. | **BASE CLOCK:** 3.8 GHz | **BOOST CLOCK:** 4.6 GHz | **PROCESS NODE:** 7nm | **L3 CACHE:** 16 MB | **L2 CACHE:** 512 KB per core | **TDP:** 65 W.

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